Paper 1 Fixes to Make the Work Publishable

This document lists the most important implementation fixes needed to make the current EchoFuseNet / EdgeFusionECGNet work strong enough for submission to at least three realistic journals, including *Biomedical Signal Processing and Control*, *Sensors*, and clinically oriented digital-health venues.[cite:4][cite:5][cite:11][cite:19][cite:21][cite:33]

Priority fixes

1. Enforce inter-patient evaluation

The most important fix is to change the current split from image-level or beat-level splitting to **patient-level inter-patient splitting** so that no patient appears in both training and test sets.[cite:19][cite:21][cite:25][cite:33]

Implementation tasks:

- Add patient IDs to the metadata for every ECG beat/image sample.
- Build DS1/DS2 style train-test partitions at the patient level.
- Use only the training patients for augmentation or oversampling.
- Keep the test set untouched and fully unseen until final evaluation.
- Report inter-patient accuracy, macro F1, weighted F1, and confusion matrix separately from the balanced internal split.[cite:19][cite:21][cite:25]

Why this matters:

- ECG papers are often criticized for data leakage when beats from the same patient appear in both train and test splits.[cite:19][cite:21][cite:33]
- Stronger journals expect inter-patient validation because it better reflects real-world deployment and generalization.[cite:21][cite:25][cite:30]

2. Make the multimodal input pipeline explicit

The code should make it completely clear that RP, GAF, and MTF are three distinct modalities loaded from different sources, rather than duplicated channels.[cite:29][cite:33]

Implementation tasks:

- Store RP, GAF, and MTF images in separate folders or indexed metadata tables.
- Update the dataset class to return `rp_img`, `gaf_img`, and `mtf_img` explicitly.
- Add assertions that the three modality files belong to the same beat and patient.
- Remove any ambiguity caused by channel replication helpers unless they are strictly needed for a later transform stage.[cite:29][cite:33]

Why this matters:

- The PhD proposal already identifies a prior channel-duplication defect as a methodological problem, so the final code must be reviewer-proof on this point.[cite:19]
- Clear modality separation improves trust in the ablation study and the paper's novelty claim.[cite:29][cite:33]

3. Add stronger external validation

The current work is more publishable if it includes at least one fully documented external validation and ideally a second external dataset.[cite:19][cite:23][cite:30][cite:33]

Implementation tasks:

- Keep MIT-BIH external validation fully reproducible in code.
- Ensure no oversampling is used on any external test set.
- Add another public dataset such as INCART or PTB-XL if possible.
- Report cross-dataset accuracy, weighted F1, macro F1, and class-wise recall.
- Include confusion matrices and short error analysis for rare classes.[cite:23][cite:30][cite:33]

Why this matters:

- External validation is one of the clearest signals that the model is not overfitted to one dataset.[cite:23][cite:30]
- It helps the paper fit both engineering journals and more clinically oriented journals.[cite:11][cite:23][cite:29]

4. Expand ablation and significance testing

The paper will be stronger if the fusion benefit is supported by repeated experiments and statistical testing rather than single-run comparisons.[cite:19][cite:21][cite:33]

Implementation tasks:

- Run patient-level k-fold cross-validation on the training cohort.
- Compare RP-only, GAF-only, MTF-only, and full-fusion models.
- Report mean ± standard deviation for accuracy and F1.
- Add statistical significance testing such as paired t-test, bootstrap confidence intervals, or McNemar testing where appropriate.[cite:19][cite:21][cite:33]

Why this matters:

- Reviewers often ask whether the fusion improvement is real or just due to one favorable split.[cite:19][cite:21]
- Significance testing makes the contribution more convincing for journals with stricter methodological review.[cite:21][cite:33]

5. Align manuscript, code, and experiment settings

The training details in the paper and the notebook must match exactly.[cite:19][cite:24][cite:33]

Implementation tasks:

- Finalize one canonical setup for batch size, epochs, learning rate, weight decay, augmentation, and early stopping.
- Make sure the manuscript reports the same values used in the final notebook.
- Save all final settings to a config file or JSON artifact.
- Log software versions, random seeds, and hardware used for training and inference.[cite:24][cite:33]

Why this matters:

- Inconsistencies between paper and code are a common reason reviewers question reproducibility.[cite:24][cite:33]
- Clear reporting increases trust and helps with revision responses later.[cite:19][cite:24]

Extra fixes that improve journal reach

6. Improve robustness analysis

The noise analysis is already useful, but it should be framed more rigorously.[cite:11][cite:19][cite:33]

Implementation tasks:

- Evaluate several noise levels with repeated runs.
- Add confidence intervals or standard deviations across runs.
- Include a brief discussion of the failure region at extreme noise.
- Optionally test baseline wander or powerline interference, not only Gaussian noise.[cite:11][cite:19][cite:33]

7. Strengthen interpretability section

The Grad-CAM section will become more convincing if it shows more representative examples and clearer clinical interpretation.[cite:23][cite:29][cite:30]

Implementation tasks:

- Show examples from each class across RP, GAF, and MTF.
- Add short class-wise interpretation comments tied to ECG morphology where possible.
- Include both correct and incorrect predictions for transparency.[cite:23][cite:29][cite:30]

8. Add deployment realism

For edge-focused journals, the deployment section should go beyond parameter count alone.[cite:8][cite:11][cite:17]

Implementation tasks:

- Benchmark inference on CPU, ONNX Runtime, and at least one edge device if available.
- Report latency, memory footprint, FPS, and model size.
- If possible, add quantization or pruning results as a small deployment extension.[cite:8][cite:11][cite:17]

Suggested order of implementation

1. Patient-level DS1/DS2 split.
2. Explicit RP/GAF/MTF loading.
3. Re-run main experiments with honest inter-patient reporting.
4. Add external validation.
5. Add k-fold ablation and significance tests.
6. Synchronize manuscript text with final code.
7. Expand Grad-CAM and deployment reporting.[cite:19][cite:21][cite:23][cite:33]

Journals this should support

If the above fixes are completed, the work should be reasonably competitive for these journals:

Journal	Why it fits
Biomedical Signal Processing and Control	Strong fit for ECG signal processing, lightweight deep learning, and deployment-focused biomedical models.[cite:4][cite:5]
Sensors	Good fit for wearable, IoMT, and edge-device ECG monitoring with latency and efficiency analysis.[cite:11][cite:13]
Frontiers in Digital Health or similar applied digital-health venue	Good fit if the paper adds stronger clinical framing, external validation, and practical deployment discussion.[cite:23][cite:29][cite:30]

Minimum publishable checklist

- Patient-level train/test split implemented.[cite:19][cite:21][cite:25]
- No leakage from oversampling or augmentation into test data.[cite:19][cite:33]
- External validation clearly reported.[cite:23][cite:30]
- Multimodal pipeline explicit and reviewer-proof.[cite:29][cite:33]
- Ablation study includes repeated runs or statistical testing.[cite:21][cite:33]
- Manuscript and code use identical settings.[cite:24][cite:33]

- Latency and model size reported for edge claims.[cite:8][cite:11][cite:17]